

ISAMBAJ, Tajikistan

WMO: 388380

Lat: 38.050N

Lon: 68.350E

Elev: 563

StdP: 94.74

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.4	-6.0	-11.4	1.5	-8.8	-8.0	2.0	-3.2	10.1	9.8	9.2	10.6	0.7	50	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	40.9	20.2	39.6	19.8	38.3	19.4	21.2	37.5	20.6	36.8	20.1	36.0	2.6	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.5	24.0	15.5	11.8	23.9	14.8	11.2	23.8	64.3	37.8	62.1	36.7	60.1	36.2	24.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.4	6.7	-12.7	44.2	6.3	1.1	-17.2	45.0	-20.9	45.7	-24.5	46.3	-29.1	47.2		
(5)			-12.4	22.9	6.3	0.9	-16.9	23.5	-20.6	24.0	-24.1	24.5	-28.7	25.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	4.2	6.7	13.0	18.0	23.5	28.8	31.2	29.3	24.7	17.9	10.3	5.2		
	DBStd	10.16	5.05	5.19	4.19	4.22	3.70	2.88	2.10	2.36	2.69	3.83	4.63	4.22		
	HDD10.0	512	182	109	19	3	0	0	0	0	0	1	46	153		
	HDD18.3	1700	437	327	169	53	5	0	0	0	1	56	246	406		
	CDD10.0	3361	3	16	113	244	418	565	658	598	441	245	54	6		
	CDD18.3	1506	0	0	5	44	165	315	399	339	192	42	4	0		
	CDH23.3	19613	0	2	56	460	1788	4215	5838	4535	2194	491	35	0		
	CDH26.7	11504	0	0	10	159	886	2553	3761	2812	1156	161	7	0		
(14) Wind	WSAvg	2.5	2.2	2.9	3.0	2.5	2.9	2.9	2.4	2.3	2.3	2.2	2.0	2.1		
(15) Precipitation	PrecAvg	652	75	94	115	103	61	19	8	4	7	26	55	80		
	PrecMax	997	141	148	198	326	130	48	18	8	28	52	159	153		
	PrecMin	410	36	24	47	33	8	6	3	2	1	11	16	24		
	PrecStd	141	29	32	36	54	36	12	3	2	6	12	33	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.8	21.4	27.9	34.4	38.1	41.6	43.6	41.8	37.6	33.1	27.6	18.4		
		MCWB	9.1	11.7	14.6	17.3	19.1	20.3	20.9	20.3	18.3	16.4	14.8	10.2		
	2%	DB	14.3	17.7	24.9	30.6	35.7	39.7	41.6	39.9	36.1	30.4	23.0	16.0		
		MCWB	8.3	10.3	14.1	16.8	18.5	19.5	20.3	20.0	17.8	15.6	12.4	9.1		
	5%	DB	12.4	15.4	22.6	28.1	33.7	38.4	40.1	38.6	34.6	28.4	20.3	13.8		
		MCWB	7.6	9.3	13.4	16.5	18.1	19.4	20.1	19.4	17.5	15.0	11.3	8.4		
	10%	DB	10.7	13.6	20.2	25.8	31.7	36.8	38.7	37.1	32.8	26.1	17.8	11.8		
		MCWB	6.9	8.6	12.6	16.0	17.6	19.1	19.7	19.0	16.9	14.2	10.5	7.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	12.8	16.4	19.7	20.8	21.6	22.4	21.7	19.9	17.3	15.0	11.5		
		MCDB	14.5	18.9	24.5	28.9	31.3	36.9	39.6	37.9	34.9	29.5	24.8	16.6		
	2%	WB	9.1	11.3	15.1	18.5	19.7	20.8	21.4	20.7	18.7	16.2	13.6	10.0		
		MCDB	12.9	15.9	22.5	27.0	31.8	36.5	38.6	37.8	33.6	28.1	21.0	13.9		
	5%	WB	8.1	10.1	14.0	17.5	19.0	20.1	20.8	20.1	17.9	15.4	12.2	8.8		
		MCDB	11.5	14.1	21.1	25.7	30.8	35.7	37.6	36.9	32.7	26.9	18.3	12.8		
	10%	WB	7.2	9.1	13.0	16.6	18.4	19.6	20.2	19.5	17.2	14.6	11.1	7.8		
		MCDB	10.1	12.7	18.7	24.1	29.2	34.6	36.8	35.6	31.6	25.3	16.5	11.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.4	11.0	12.0	13.8	14.9	15.3	15.7	16.1	14.7	11.4	9.9		
		MCDBR	11.5	12.9	14.3	15.9	16.5	16.4	16.7	16.7	17.4	17.2	15.9	13.9		
	5% WB	MCWBR	6.5	6.4	6.2	5.8	4.6	4.7	4.5	4.6	5.4	6.5	7.4	8.0		
		MCWBR	9.9	10.8	12.5	13.8	14.5	15.3	15.4	16.0	16.6	15.7	13.1	11.5		
(40) Clear-Sky Solar Irradiance	taub	taub	0.387	0.440	0.501	0.506	0.512	0.541	0.535	0.502	0.467	0.495	0.452	0.400		
		taud	2.095	1.966	1.850	1.875	1.890	1.815	1.821	1.888	1.932	1.837	1.926	2.039		
	Ebn at Noon	Ebn at Noon	755	759	753	781	785	760	760	774	774	689	665	700		
		Edh at Noon	118	153	190	196	196	212	209	191	172	172	139	117		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.06	2.74	4.00	5.39	6.68	7.74	7.78	7.17	5.97	4.13	2.51	1.86		
	RadStd	RadStd	0.25	0.39	0.37	0.53	0.44	0.31	0.21	0.23	0.17	0.34	0.30	0.26		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (10 neighbors)	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	-0.36	-0.68	N/A	N/A	N/A

Nomenclature: See separate page